J. Electroanal. Chem., 146 (1983) 263–278
Elsevier Sequoia S.A., Lausanne – Printed in The Netherlands

A DC, AC AND DEMODULATION STUDY OF THE TRIS-OXALATO Fe(III)/TRIS-OXALATO Fe(II) REDOX COUPLE IN 1 *M* AQUEOUS OXALATE SOLUTION AT THE DROPPING MERCURY ELECTRODE

J. STRUIJS, M. SLUYTERS-REHBACH and J.H. SLUYTERS

Van 't Hoff Laboratory of Physical and Colloid Chemistry, State University, Padualaan 8, 3584 CH Utrecht, (The Netherlands)

(Received 28th June 1982)

ABSTRACT

The reduction of the Fe(III) oxalate complex at the dropping mercury electrode in a solution of 1 *M* $K_2C_2O_4$ + 0.05 *M* $H_2C_2O_4$ is studied, combining the results of dc, ac and demodulation polarography. It is shown that the double-layer capacity is influenced by the presence of the reactants due to their adsorption. A careful analysis is undertaken to explain the extreme value of the transfer coefficient. Contrary to earlier beliefs, the results indicate that the kinetics do not follow simple Butler–Volmer behaviour, but rather should be described in terms of an EC mechanism (electron transfer followed by a heterogeneous chemical step). The kinetic parameters are reported and compared with data in the literature.

(I) INTRODUCTION

Recently we presented a theoretical treatment of the use of demodulation polarography for kinetic studies of electrode processes [1]. In a subsequent paper the experimental set-up was described in detail and the proper working of the demodulation polarograph was demonstrated in respect of the charging process of a mercury electrode in contact with 1 *M* KCl and 1 *M* $K_2C_2O_4$ + 0.05 *M* $H_2C_2O_4$ respectively [2].

The ferric–ferrous electrode reaction in oxalate medium appeared to be a suitable object to verify a new technique for investigations of fast electrode processes, as the literature values of the kinetic parameters obtained with different methods show good agreement [3–7]. It was concluded by some investigators that there is no irregular behaviour, as both faradaic rectification [3,4] and the galvanostatic double-pulse method [5,6] give quite similar results. However, the observed cathodic transfer coefficients, with values between 0.8 and 0.9 are quite extreme, and this gives rise to some suspicion whether the Butler–Volmer theory describes this redox couple sufficiently well because such an asymmetrical potential dependence is not plausible.

In the present paper, applying the demodulation method [1,2], the standard heterogeneous reaction constant is found to be slightly higher than that obtained by

0022-0728/83/$03.00

the authors quoted. The transfer coefficient, however, is found to be even more extreme (almost unity), indicating that a fast electrochemical step followed by a rate-determining heterogeneous chemical reaction is a more likely reaction mechanism.

An electrode process in this range of reversibility can, although less accurately, be also studied with the faradaic impedance method, provided that complications, involving the electrical double-layer structure, do not interfere. However, the Fe(III) and Fe(II) oxalato complexes are highly charged and should therefore be expected to influence the value of the double-layer capacity. By careful analysis, combining the results of demodulation and impedance measurements, it will be shown that not taking this influence into account leads to erroneous conclusions.

(II) THEORY

If Randles behaviour prevails, the first- and second-order behaviour of the interface may be described in terms of the parameters collected in Table 1. The meaning of those parameters was given previously [1] in an implicit way, i.e. without adopting a kinetic model, but only stating that the *faradaic current density* j_F is a function of the interfacial potential E and the surface concentrations c_O and c_R, while the *non-faradaic current density* j_C is merely a function of the potential E. In view of the analysis to be applied, we have specified in Table 1 the potential dependence of the several parameters for the case that the faradaic reaction is of first-order stoichiometry, obeys the absolute rate theory and is dc reversible.

The way in which the several parameters occur in the experimental quantities will be briefly reviewed in the following subsections.

TABLE 1

Parameters used in eqns. (3) and (4) and their potential dependence, assuming linear kinetics, Butler–Volmer theory and dc reversibility

Symbol	Meaning
R_{ct}	$(RT/nF[nFk_f c_O]^{-1}$
R'_{ct}	$-(RT/nF)^2[nFk_f c_O(1-2\alpha)]^{-1}$
λ	$k_f[D_O^{-1/2} + D_R^{-1/2}\exp(\phi)]$
λ'_f	$-k_f[\alpha D_O^{-1/2} - (1-\alpha)D_R^{-1/2}\exp(\phi)]$
λ'_c	0
p_H	$(2\omega_H)^{1/2}k_f^{-1}[D_O^{-1/2} + D_R^{-1/2}\exp(\phi)]^{-1}$
a_H	$(\omega_H/2)^{-1/2}c_d(n^2F^2/RT)c_O[D_O^{-1/2} + D_R^{-1/2}\exp(\phi)]^{-1}$

(II.1) Demodulation polarography

In view of the reasonings given previously [1,2] the perturbing ac signal is of the form

$$j = j_A \sin \omega_H t \cos \omega_L t \tag{1}$$

in which j_A is the amplitude of the total alternating current flowing through the cell. The detection of the voltage response has to be carried out at the frequency $2\omega_L$.

Figure 1 shows the equivalent circuit containing the "virtual" second-order voltage sources S_c and S_F due to the charging process and the faradaic process respectively. An expression for the demodulation voltage ΔE_M can be derived according to Kirchhoff's rules. This voltage has an in-phase I_{LF} and a quadrature component Q_{LF} as has been derived earlier [1]:

$$I_{LF} = [S_c] - ([S_c] - [S_F]) a_L (a_L + 1)/\left[(p_L + 1)^2 + (a_L + 1)^2\right] \tag{2a}$$

$$Q_{LF} = ([S_c] - [S_F]) a_L (p_L + 1)/\left[(p_L + 1)^2 + (a_L + 1)^2\right] \tag{2b}$$

where $p_L = (4\omega_L)^{1/2} \lambda^{-1}$ and $a_L = (2\omega_L^{1/2}\sigma C_d)^{-1}$ are the parameters determining the interfacial admittance at the frequency $2\omega_L$.

The expressions derived for $[S_F]$ and $[S_c]$ in terms of the parameters in Table 1 are [1]

$$[S_F] = \left\{ -\frac{R_{ct}}{2R'_{ct}}\left(p_H^2 + 2p_H + 2\right) + \frac{nF}{RT}\frac{\lambda'_f}{\lambda}(p_H + 2) + \frac{\lambda'_c}{\lambda^2 R_{ct}} \right\} \times \frac{(j_A/2\omega_H C_d)^2}{(p_H + 1)^2 + (a_H + 1)^2} \tag{3}$$

$$[S_c] = -\frac{1}{2C_d}\frac{dC_d}{dE}\frac{(j_A/2\omega_H C_d)^2\{(p_H + 1)^2 + 1\}}{(p_H + 1)^2 + (a_H + 1)^2} \tag{4}$$

where $p_H = (2\omega_H)^{1/2} \lambda^{-1}$ and $a_H = (\omega_H^{1/2}\sigma C_d)^{-1}$.

In eqns. (2) the influence of the external circuit on the detected demodulation voltage has not been taken into account. In the instrumental part of the series [2], it was explained how the admittance Y_{ex} of the external circuit (in our case $Y_{ex} = i2\omega_L C_{ex}$, where C_{ex} is the sum of four capacitances) has to be incorporated in a formal way. Making eqn. (8) of ref. 2 explicit in terms of p_L and a_L enables us to write the real (in-phase) component I_{LF} and the imaginary (quadrature) component

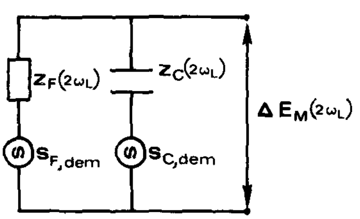

Fig. 1. Equivalent circuit of the electrochemical cell connected to a high-input impedance across which the response $\Delta E_M(2\omega_L)$ is measured.

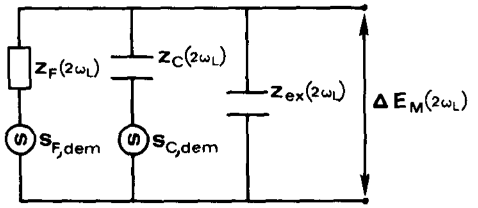

Fig. 2. Equivalent circuit and the influence of the measuring network.

Q_{LF} as follows:

$$I_{LF} = \frac{[S_c]}{1+X_c} - \left(\frac{[S_c]}{1+X_c} - [S_F] \right) \frac{a_L^\star (a_L^\star + 1)}{(p_L + 1)^2 + (a_L^\star + 1)^2} \tag{5a}$$

$$Q_{LF} = \left(\frac{[S_c]}{1+X_c} - [S_F] \right) \frac{a_L^\star (p_L + 1)}{(p_L + 1)^2 + (a_L^\star + 1)^2} \tag{5b}$$

where $a_L^\star = a_L/(1+X_c)$, $X_c = A^{-1}C_{ex}/C_d$ and A is the electrode surface area. Therefore, eqns. (5) are to be applied in practice, if the experimental set-up has in principle the conformation as schematized in Fig. 2.

(II.2) Faradaic admittance method

The interfacial admittance Y can be computed from impedance data after subtraction of the ohmic resistance. If the rate of the electrode reaction is controlled only by diffusion and electron transfer, the real component Y' and the imaginary component Y'' of the interfacial admittance are consistent with the expressions [8]

$$Y' = (\omega^{1/2}/\sigma)(p+1)/\left[(p+1)^2 + 1\right] \tag{6a}$$

$$Y'' = (\omega^{1/2}/\sigma)\left[(p+1)^2 + 1\right]^{-1} + \omega C_d \tag{6b}$$

$$= (\omega^{1/2}/\sigma)\left\{\left[(p+1)^2 + 1\right]^{-1} + 1/a\right\} \tag{6c}$$

with $p = p'\omega^{1/2} = (2\omega)^{1/2}/\lambda$ and $a = (\omega^{1/2}\sigma C_d)^{-1}$. Note that $p'\sigma = R_{ct}$. Application of the frequency variation method [8] yields values for the charge-transfer resistance R_{ct}, the Warburg coefficient σ and the double-layer capacity. These quantities can be obtained by fitting to the $\omega^{1/2}/Y'$ vs. $\omega^{1/2}$ and $\omega^{1/2}/(Y'' - \omega C_d)$ vs. $\omega^{1/2}$ plots. Normal "Randles behaviour" then implies that C_d will be equal to the value obtained in the supporting electrolyte alone at corresponding potentials.

As usual, eqn. (6a) applied at different frequencies, gives a pair of σ and p' values which best fit the frequency dispersion of $\omega^{1/2}/Y'$. However, if charge transfer is too fast, only σ can be obtained because p' approximates zero. The lowest value of p' to be determined is $p' = 10^{-3}$ $s^{1/2}$ [9], provided that the proper concentration is chosen to adjust the optimal σ value. In ref. 9 it has been argued that eqn. (6b) is more sensitive to small p' values ($p' = 0.2 \times 10^{-3}$ $s^{-1/2}$), although its accuracy is somewhat less. However, fitting on $\omega^{1/2}/(Y'' - \omega C_d)$ might give erroneous results, because it requires a priori knowledge of the double-layer capacity, which is not a priori equal to that measured in the pure base electrolyte. When p' is small, a slight discrepancy between the two will lead to a large error in the p' value obtained from the fitting analysis.

(II.3) Combination of demodulation and admittance measurements

As previously pointed out, a useful relationship can be derived by eliminating $[S_c]$ between I_{LF} and Q_{LF} [1]. If this is applied to eqns. (5a) and (5b), one obtains

$$I_{LF} - Q_{LF}\{1 + [(p_L + 1)^2 + 1]/a_L^\star\}/(p_L + 1) = [S_F] \tag{7}$$

and, in view of eqns. (6) it is easily verified that this is identical to

$$I_{LF} - Q_{LF}(Y_L'' + Y_{ex}'')/Y_L' = [S_F] \tag{8}$$

where Y_L' and Y_L'' are the real and imaginary components of the interfacial admittance *measured at the frequency* $2\omega_L$, and $Y_{ex}'' = 2\omega_L C_{ex} A^{-1}$.

(III) EXPERIMENTAL

All measurements were made at 25.0 ± 0.1°C in a three-electrode cell, with a dropping mercury electrode (DME), a mercury pool counter electrode and a saturated calomel electrode (SCE) as a reference electrode. Deaeration was performed with purified argon. The solutions were made from analytical grade reagents dissolved in twice distilled water.

(III.1) Faradaic admittance method

The amplitude and phase angle of the impedance of the cell were measured as a function of frequency and dc potential, using the automatic network analyser system [10]. The working electrode was a finely drawn-out capillary of about 0.3 mm diameter to avoid shielding as much as possible. The natural drop time was ca. 6 s. Measurements were taken at 4 s after drop birth.

(III.2) Demodulation method

The circuit has been described previously [2]. Unless stated otherwise, all experimental details were completely analogous to those in ref. 2. The working electrode was a normal blunt capillary.

The demodulation signals were kept below 1 mV. Higher-order effects arising from too large an amplitude of the perturbing ac current are expected to have only a negligible contribution under this condition [11]. In order to check this, the influence

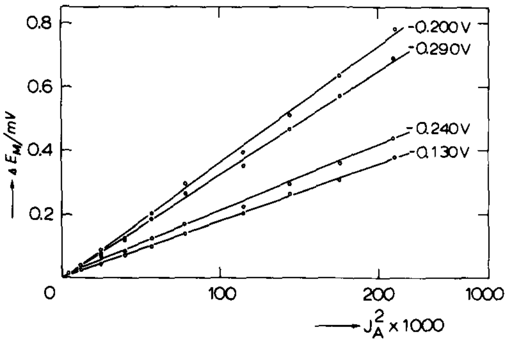

Fig. 3. The amplitude of the demodulation response as a function of the squared amplitude of the total high-frequency perturbating ac current. Measured with electrode area 0.025 cm^2; $f_H = 100$ kHz, $f_L = 140$ Hz, $c^*_{Fe^{3+}} = 1$ mM.

of a fourth-order contribution, which would be the most severe, was checked on its absence. To this end, two criteria were examined: Firstly, the absence of a signal at the frequency $4\omega_L$ was verified using a spectrum analyser. Secondly, the linear relationship between the amplitude of the demodulation signal and the squared amplitude of the current density was confirmed by experiments shown in Fig. 3. The results clearly show that any contribution due to fourth-order non-linearity must be negligible.

(IV) RESULTS

From limiting currents of dc polarograms the following values for the diffusion coefficients were calculated: $D_O = D_R = 5.2 \times 10^{-6}$ cm^2 s^{-1}. This value is somewhat higher than those previously reported [3–7]. The half-wave potential was -0.231 ± 0.002 V vs. SCE.

(IV.1) Preliminary demodulation measurements

The amplitude $|\Delta E_m| = (I^2_{LF} + Q^2_{LF})^{1/2}$ was measured at dc potentials between 0.0 and -0.5 V vs. SCE. Figure 4a, b shows these demodulation polarograms at 100 and 200 kHz respectively of a solution containing 1 mM Fe^{3+}, 1 M $K_2C_2O_4$ and 0.05 M $H_2C_2O_4$. The modulation frequency was 142 Hz, so the detection was carried out at the doubled frequency: 284 Hz. The solid line in Fig. 4 was calculated according to eqns. (3)–(5), assuming Butler–Volmer kinetics apply. The best fit was obtained with the following set of kinetic parameters:

$\alpha = 1.0 \qquad k_{sh} = 1.6$ cm s^{-1} at $f_H = 100$ kHz

$\alpha = 1.0 \qquad k_{sh} = 1.5$ cm s^{-1} at $f_H = 200$ kHz

In accordance with what we have shown recently [1] for electrode processes with a rate in this order of magnitude and an extreme transfer coefficient, the demodula-

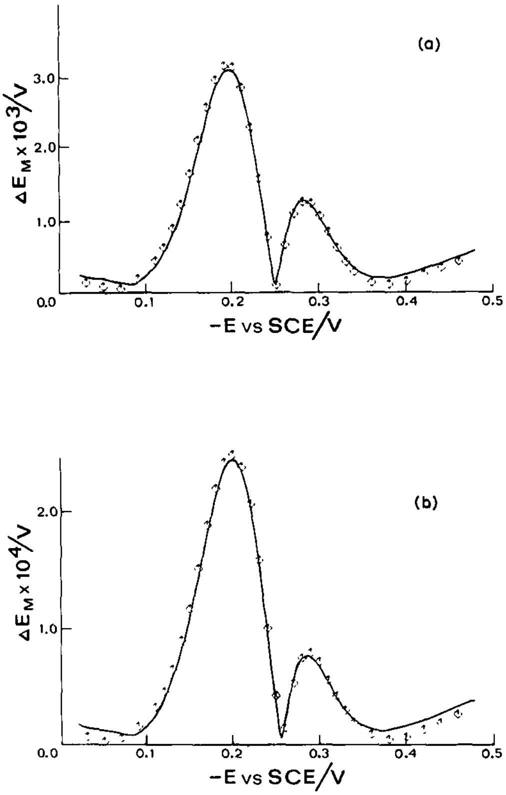

Fig. 4. Demodulation polarograms of 1 mM Fe^{3+} in 1 M $K_2C_2O_4$, 0.05 M $H_2C_2O_4$; $f_L = 142$ Hz, $f_H = 100$ kHz (a), 200 kHz (b); $j_A = 0.1408$ A cm^{-2} (a), 0.2346 A cm^{-2} (b).

tion polarogram at 200 kHz (Fig. 4b) has a more asymmetric shape than that at 100 kHz (Fig. 4a).

(IV.2) Phase-sensitive demodulation measurements

At 50 dc potentials the in-phase and quadrature were measured (Fig. 5a, b) at 340 Hz. The drawn lines correspond to the following data:

in-phase: $\alpha = 1.0$ $\quad k_{sh} = 1.65$ cm s^{-1}

quadrature: $\alpha = 1.0$ $\quad k_{sh} = 1.64$ cm s^{-1}

Both components of the response ΔE_M show a good consistency, and again a relative high value of k_{sh} and a most extreme α gave the best fit, although at potentials more negative than -0.280 V vs. SCE there is a slight discrepancy. Both Figs. 4 and 5 were obtained with values of $[S_c]$ calculated from double-layer data pertaining to the pure base electrolyte [2].

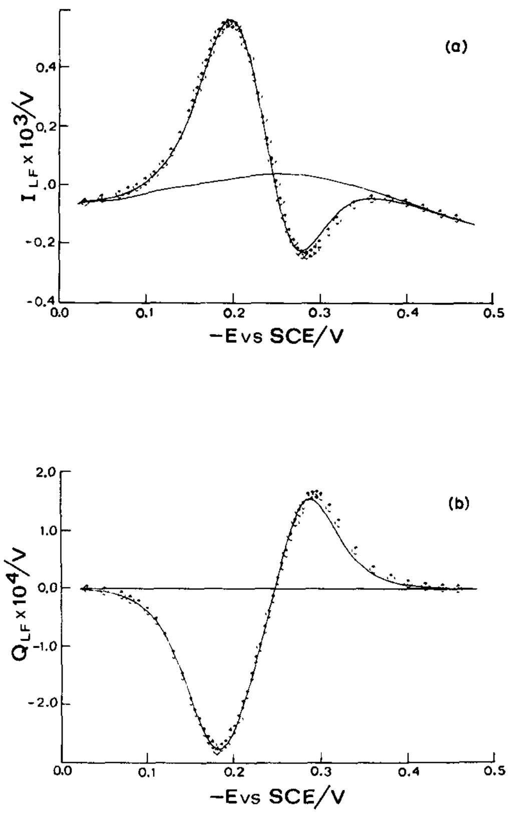

Fig. 5. In-phase (a) and quadrature (b) components of ΔE_m; $f_H = 100$ kHz, $f_L = 170$ Hz, $j_A = 0.2089$ A cm^{-2}.

(IV.3) Faradaic admittance measurements

If the irreversibility quotient p' were expressed in terms of Butler–Volmer kinetics it could be possible to obtain the kinetic parameters of the ferric–ferrous couple. With values of k_{sh} and α, as collected in Table 2, it can easily be computed that in the faradaic region we have $0 < p' < 0.003$ s^{-1}. When $D_O = 5.2 \times 10^{-6}$ cm^2 s^{-1} is inserted, this follows from eqn. (9):

$$p' = (2D_O)^{1/2} k_f^{-1} \left[1 + (D_O/D_R)^{1/2} \exp(\phi)\right]^{-1} \tag{9}$$

with $k_f = k_{sh} \exp(-\alpha\phi)$ and $\phi = (nF/RT)(E - E^0)$ [1,9,10].

In view of the foregoing, analysis of the frequency dispersion of the quadrature component of the interfacial admittance should at least give some information about the kinetics of the Fe(III)/Fe(II) system if the Randles equivalent circuit applies.

Plots of $\omega^{1/2}/Y'$ vs. $\omega^{1/2}$ for 1 mM and 5 mM Fe(III) solutions, gave fairly

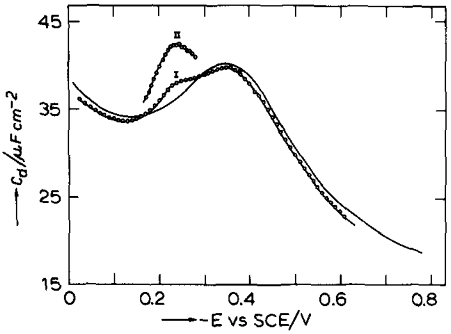

Fig. 6. Capacity–potential curves of 1 M $K_2C_2O_4$, 0.05 M $H_2C_2O_4$ (line), in the presence of 1 mM Fe^{3+} (curve I) and 5 mM Fe^{3+} (curve II).

horizontal straight lines (i.e. $p' \simeq 0$) at each dc potential, from which the Warburg coefficient σ was obtained directly. The potential dependence of σ was as expected, consistent with a diffusion coefficient $D_O = 5.2 \times 10^{-6}$ cm^2 s^{-1}. However, from the $\omega^{1/2}/(Y'' - \omega C_d)$ vs. $\omega^{1/2}$ analysis remarkable results emerged. At potentials negative with respect to the half-wave potential, p' appeared to be negative and this effect increased on going to more cathodic potentials. Neither a three-parameter fit (on σ, p' and C_d), nor a fit on p' and C_d with substitution of a fixed σ value [obtained from Y'] led to an acceptable result. However, all these attempts strongly suggested the conclusion that the presence of the ferric and/or ferrous oxalato complexes influences the value of the double-layer capacity. Careful determination of the double-layer capacity in the *non-faradaic regions* ($0 > E > -0.1$ V vs. SCE and $-0.35 > E > -0.6$ V vs. SCE), but in the presence of 1 mM Fe^{3+}, confirmed this idea, as can be seen from the suppression of C_d shown in Fig. 6.

(IV.4) Combination of demodulation and admittance results

Considering that the kinetic results from the demodulation experiments are much less sensitive to contributions from double-layer charging, we thought it worth while to attempt a possibly somewhat revolutionary way of analysis of the admittance data: instead of inserting a priori C_d values, we assumed the data in section (IV.2) to be correct and thus calculated p' as a function of potential with $D_O = 5.2 \times 10^{-6}$ cm^2 s^{-1}, $\alpha = 1.0$ and $k_{sh} = 1.64$ inserted into eqn. (9). Next, these p' values together with σ values obtained from Y' were introduced into eqn. (6b), whereupon it was possible to compute C_d as a function of dc potential from the experimental Y'' data, by a one-parameter fit.

The results of this exercise, given in Fig. 6, demonstrate that the C_d vs. E curves in the faradaic region deviate from the base-electrolyte data (drawn curve) appreciably in the presence of 1 mM Fe(III) and dramatically in the presence of 5 mM Fe(III). It should be noted that with lower values for the kinetic parameters to calculate p', e.g. $\alpha = 0.9$ and $k_{sh} = 1.3$ cm s^{-1} (see Table 2), the resulting C_d would be even more enhanced in comparison to the "normal" data. This excludes the possibility that the enhancement of the double-layer capacitance was a mathematical artefact.

Naturally, the next step is to check whether the new capacity data, if used in the demodulation expressions, influence the kinetic results. Therefore, we recalculated the function $[S_c]$, given by eqn. (4), using a 10th-degree polynomial fitted through the curve for 1 mM Fe(III) in Fig. 6 to calculate dC_d/dE. Also, the values of a_H in eqns. (3) and (4) and $a_L^\star$ in eqn. (5) were re-estimated. Thereupon, we found that the experimental data for I_{LF} and Q_{LF} gave a best fit to eqn. (5) with the next set of kinetic parameters

in-phase: $\alpha = 1.0$ $\quad k_{sh} = 1.52$ cm s^{-1}

quadrature: $\alpha = 1.0$ $\quad k_{sh} = 1.48$ cm s^{-1}

Note that the concordance of the data with the calculated curves (see Fig. 7) is even better than in Fig. 5. The slight change of only the rate constant justifies the renunciation of further iterations.

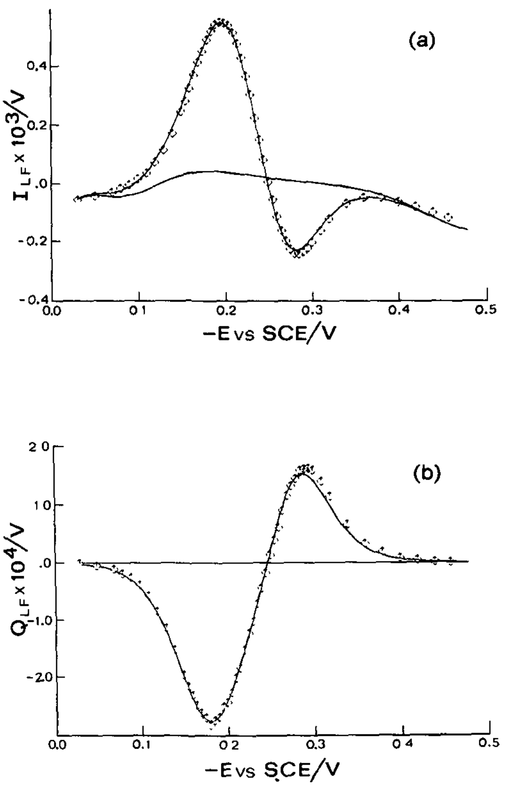

Fig. 7a,b. As Fig. 5. For explanation, see text.

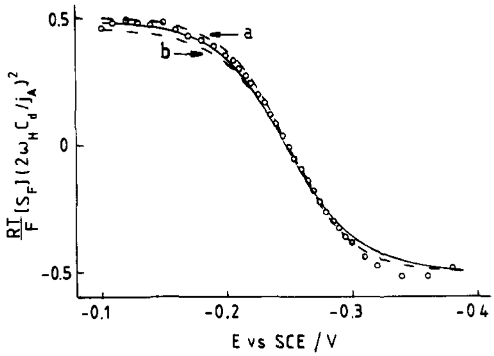

Fig. 8. Comparison of theoretical and experimental values of $[S_F]$. (○) obtained using eqn. (8). (— — —) Calculated with eqn. (3): (a) $\alpha = 1.0$, $k_{sh} = 1.50$ cm s^{-1}; (b) $\alpha = 0.9$, $k_{sh} = 1.35$ cm s^{-1}. (———). Calculated with eqn. (10): $\alpha_1 = 0.5$, $k_{s1} = 8$ cm s^{-1}, $k_c K_c = 1.6$ cm s^{-1}.

A quite valuable check on both the correctness of our procedure and the precision of the results can be performed by application of eqn. (8) in section (II.3). Particularly profitable in the present case is the fact that there is no need to split the imaginary admittance component Y_L'' into the contributions from the faradaic admittance and the double-layer capacity. It is just the raw data that have to be inserted to calculate $[S_F]$. The result is plotted as $(RT/F)[S_F](2\omega_H C_d/j_A)^2$ vs. potential in Fig. 8 (○) together with two theoretical curves, calculated with eqn. (3) for the following combinations of α and k_{sh}:

(a) $\alpha = 1.0$ $\quad k_{sh} = 1.5$ cm s^{-1}
(b) $\alpha = 0.9$ $\quad k_{sh} = 1.35$ cm s^{-1} $\quad (D_O = 5.2 \times 10^{-6}$ cm^2 s$^{-1})$

The theoretical curves were hardly sensitive to a change of 0.15 in k_{sh}. The α value appears to be closer to 1.0 than to 0.9, but in view of the experimental accuracy the final estimate of the kinetic parameters is

$\alpha = 0.97 \pm 0.03 \qquad k_{sh} = 1.5 \pm 0.15$ cm s^{-1}

(IV.5) The possibility of a two-step mechanism

Although within the experimental accuracy Butler–Volmer kinetics, i.e.

$$k_f = k_{sh} \exp[-\alpha(nF/RT)(E - E^0)]$$

appears to be obeyed, the extreme value of the transfer coefficient is puzzling. As it means that the rate of oxidation will be independent of potential, it should not be excluded that a more complex reaction scheme is at hand, involving a non-electrochemical step as follows:

$$\mathrm{O} + e \underset{k_1 K_1}{\overset{k_1}{\rightleftarrows}} \mathrm{Y}$$

$$\mathrm{Y} \underset{k_c K_c}{\overset{k_c}{\rightleftarrows}} \mathrm{R}$$

If k_c and K_c are independent of potential and k_1 is of the form $k_1 = k_{s1} \exp[-\alpha_1 \phi]$, where $\phi = (F/RT)(E - E^0)$(with $\alpha_1 \simeq 0.5$), the overall forward rate constant of reduction k_f will be given by

$$k_f^{-1} = k_{s1}^{-1} \exp[\alpha_1 \phi] + (k_c K_c)^{-1} \exp(\phi) \tag{10}$$

Evidently the second term must be dominating, since the overall operational transfer coefficient $\alpha = -\mathrm{d} \ln k_f/\mathrm{d}\phi$ is found close to unity. In agreement with the model underlying eqn. (10), a three-parameter fit [on α_1, k_{s1} and $k_c K_c$] was applied to the experimental data I_{LF} and Q_{LF}. This gave the following results:

$\alpha_1 = 0.5–0.6$

$k_{s1} = 8–4 \quad \mathrm{cm\ s^{-1}}$

$k_c K_c = 1.6–1.9\ \mathrm{cm\ s^{-1}}$

If with these sets of parameters k_f is reconstructed, it appears in Fig. 9 that a slight curvature is introduced into the plot of ln k_f vs. potential. The corresponding values of α change remarkably strongly with potential: from 0.99 at $E - E^0 = +0.12$ V via 0.88 at $E = E^0$ to 0.7 at $E - E^0 = -0.12$ V. The corresponding value of k_f at $E = E^0$ is $k_f(E^0) = 1.35\ \mathrm{cm\ s^{-1}}$.

The theoretical $[S_F]$ vs. E curve, calculated with the data above, exactly concords with the data points in Fig. 8, except for those at $E < -0.3$ V which evidently are

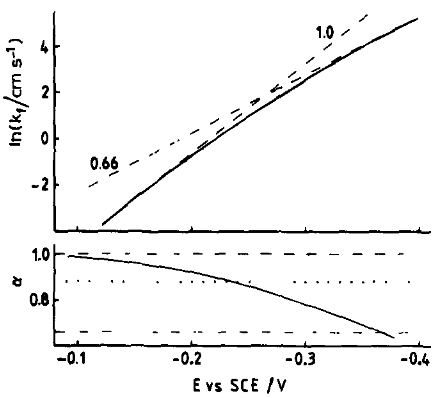

Fig. 9. The forward rate constant of reduction and the operational transfer coefficient as a function of potential. (— — —). Slopes and α-values (indicated) corresponding to the limits of the experimental potential range; (· · · · · ·) α-value at $E = E_{1/2}$; (———) calculated with eqn. (10) with the same data as in Fig. 8.

affected by a systematic error, because always [1]

$$-0.5 < (RT/nF)[S_F](2\omega_H C_d/j_A)^2 < 0.5$$

(V) DISCUSSION

(V.1) The kinetics of charge transfer

The Fe(III)/Fe(II) redox couple in 1 M oxalate solution has often served as an example of a simple fast one-electron transfer reaction that is most suitable to test an instrumental set-up. The results from studies available in literature are summarized in Table 2. With a view to our present study some new remarks can be made.

(1) At first sight the k_{sh} values are fairly scattered. Possibly this can be explained at least partly by the fact that the best-fitting result is strongly linked with the value inserted for the diffusion coefficient, as the ratio $D_O^{1/2}/k_{sh}$ appears in the primary parameter p' (or λ). If this is taken into account, it can be concluded that an unusually good agreement between the results obtained with different relaxation techniques, and partly in different laboratories, exists.

(2) For the discrepancies between the reported α values, a different reason has to be found. Considering our own data, it is easily seen that, adopting the Butler–Volmer model, the extreme value is obtained due to the fact that experimental data at very positive potential could be measured with quite good accuracy. On the other hand, in other second-order studies using the faradaic rectification principle [3,4], frequencies up to 1 MHz were employed, possibly guaranteeing a better sensitivity for α, but

TABLE 2

Kinetic parameters of the Fe(III)/Fe(II) electrode reaction in 1 M oxalate base electrolyte

	$k_{sh}{}^a$/ cm s^{-1}	α^b	D_O/ cm^2 s^{-1}	Potential range $(E-E^0)$/V
Ac polarography [7] (0.5 M K_2Ox)	1.2	> 0.5	?	
GDP [5]	1.44	0.85	4.8×10^{-6}	+0.06 to −0.02
GDPP [6]	1.15	0.78	?	+0.04 to −0.04
FR at equilibrium [3] (0.5 M K_2Ox + 0.05 M H_2Ox)	1.2	0.86	4×10^{-6}	−0.02 to −0.06
FR polarography [4] (0.5 M K_2Ox + 0.05 M H_2Ox)	1.3	0.84	4.8×10^{-6}	+0.08 to −0.08
Present work (BV) (0.5 M K_2Ox + 0.05 M H_2Ox)	1.5	0.97	5.2×10^{-6}	+0.120 to −0.120
Present work, two-step (0.5 M K_2Ox + 0.05 M H_2Ox)	1.35	0.88	5.2×10^{-6}	+0.120 to −0.120

[a] Defined as $k_f(E^0)$

[b] Defined as $-\mathrm{d}\ln k_f/\mathrm{d}\phi$ at $E=E^0$.

in a potential region closer to the half-wave potential. Comparing our results with those of refs. 3 and 4, we think that together they support the idea that the operational transfer coefficient is a function of the dc potential as a result of the two-step mechanism considered in section (IV.5). It would be most interesting to investigate the actual nature of the non-electrochemical activation controlled step, evidently involving the ferrous oxalate complex.

An additional observation has to be made concerning the fact that our demodulation polarograms could be fitted most acceptably to both the Butler–Volmer equation and the equation for the two-step mechanism. This is at once satisfying and alarming, because is shows that one should be extremely careful in using computerized fitting procedures in this type of work. In our case, the origin of the problem lies in the similarity of the two terms in eqn. (10). We have met analogous problems earlier while fitting admittance data to an equivalent circuit containing two Warburg-like branches in parallel [12].

(V.2) Reactant adsorption

The internal consistency of the kinetic results gives confidence as to the reliability of the double-layer capacities determined by insertion of these results into the admittance data. In our opinion this is a nice example of the merits of combination of second- and first-order ac techniques.

The observed influence of the iron-oxalate complexes on C_d needs some further discussion, as evidently we are dealing with an electrode reaction accompanied by adsorption of one or both of the reactants. It is well known [8] that in this case it is not correct to adopt the Randles equivalent circuit, as has been done in our analysis. However, the exact mathematical description including reactant adsorption as well as a finite rate of charge transfer is most complex [13,14] and will not be invoked here. The way in which the problem is analysed here is reasonable in view of the high value of the rate constant, making the system almost "ac reversible" at the low frequency of detection [340 Hz], and in view of the small effect of the reactant adsorption (C_d is increased at maximum 1.5 μF cm^{-2} at 1 mM Fe(III)) still allowing adoption of the Randles circuit with only a concentration-dependent "low-frequency capacity" [8].

The results of the present treatment resemble in a way those of the admittance analysis of the reduction of Ba^{2+} from 0.1 M LiCl solution [15] which was intended to show the phenomenon of reactant adsorption also to be of practical importance in the case of pure electrostatic adsorption in the diffuse double layer. It is tempting to think of the present electrode reaction between the highly charged $Fe(Oxalate)_3^{3-}$ and $Fe(Oxalate)_3^{4-}$ species as a better example. However, further work would be needed to know whether specific adsorption can be excluded. The potential of zero charge in the 1 M oxalate base electrolyte is -0.500 ± 0.003 V vs. SCE, 30 mV more negative than in F^- solution [16], indicating weak specific adsorption of the oxalate anion, so it is not impossible that the iron complexes are also specifically adsorbed, induced by the anions.

(V.3) The nature of the chemical step

As the number of ligands does not change in the overall process, the chemical reaction found to proceed next to the electron transfer reaction should (one imagines) to consist of internal reorganization of the oxalate ligands, the ionic self-atmosphere and the solvent sheath of the ion, part of which is shared with the surface of the metal electrode. This reorganization occurs due to the change of the charge of the central ion and should be looked upon as a relaxation to a new equilibrium state.

Theories on the rates of electrode reactions along these lines are available [17]. Their test until now has been somewhat restricted, almost without exception, to rather slow reactions [18]. There evidently is a fundamental problem in this work, that is designed to show experimentally the potential dependence of the apparent transfer coefficient. On the one hand, one wishes the electron transfer to be fast in order for the reorganization to contribute sufficiently to the overall rate to allow its study. This suggests the choice of a fast electron transfer reaction. For such reversible reactions, however, diffusion polarization contributes at small deviation from the equilibrium potential.

This severe restriction of the available potential range makes it unlikely that any potential dependency of the apparent transfer coefficient of fast reactions can be seen.

On the other hand, however, as argued above, choosing an irreversible reaction implies domination of slow electron transfer.

We think that at present measurement precision has developed to such an extent that both of these conflicting conditions can be met with the ac technique, where slow diffusion can be dealt with quantatively with a high accuracy.

A theory of electrode reactions including the reorganization of the environment of the reactant has been given by Marcus [19], Hush [20] and Sutin [21]. It seems to be generally accepted that (abandoning the ϕ_2 correction) the reorganization work λ should be incorporated in the free energy $\Delta F^\star$ of activation according to

$$\Delta F^\star = \lambda/4 + F(E - E^0)/2 + F^2(E - E^0)^2/4\lambda$$

whence

$$\alpha_{\text{app}} = \partial\Delta F^\star/F\partial(E - E^0) = \tfrac{1}{2} + F(E - E^0)/2\lambda$$

This suggests a linear dependence of α_{app} on potential, α_{app} being close to 0.5 near the standard potential. Neither expectations are in agreement with the results obtained in this work.

We therefore believe that at least for this redox couple a better description of the kinetics is a simple EC mechanism. This implies that the unrelaxed ion, i.e the ferrous oxalate ion surrounded by ions and water in a configuration still belonging to the ferric ion, should be looked upon as an intermediate, the concentration of which can be eliminated between the two rate equations of the elementary acts in order to obtain the rate equation describing the overall process.

Further work on fast one-electron transfer will be necessary in order to decide which view is the most satisfactory.

REFERENCES

1 J. Struijs, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 143 (1983) 37.
2 J. Struijs, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 143 (1983) 61.
3 R. de Leeuwe, M. Sluyters-Rehbach and J.H. Sluyters, Electrochim. Acta, 12 (1967) 1593; 14 (1969) 1183.
4 F. van der Pol, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 45 (1973) 377.
5 T. Rohko, M. Kogoma and S. Aoyagui, J. Electroanal. Chem., 38 (1972) 45.
6 H. Mizota, H. Matsuda, Y. Kanzaki and S. Aoyagui, J. Electroanal. Chem., 45 (1973) 385.
7 B.J. Huebert and D.E. Smith, Anal. Chem., 44 (1972) 1179.
8 M. Sluyters-Rehbach and J.H. Sluyters in A.J. Bard (Ed.), Electroanalytical Chemistry, Vol. 4, Marcel Dekker, New York, 1970.
9 C.P.M. Bongenaar, Thesis, Utrecht, 1980.
10 C.P.M. Bongenaar, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 109 (1980) 23.
11 D.J. Kooyman, M. Sluyters-Rehbach and J.H. Sluyters, Electrochim. Acta, 11 (1966) 1197.
12 E. Fatas, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 136 (1982) 59.
13 R.L. Birke, J. Electroanal. Chem., 33 (1971) 201.
14 H. Moreira and R. de Levie, J. Electroanal. Chem., 35 (1972) 103.
15 B. Timmer, M. Sluyters-Rehbach and J.H. Sluyters, J. Electroanal. Chem., 24 (1970) 287.
16 P. Delahay, Double Layer and Electrode Kinetics, Interscience, New York, 1965.
17 For reviews, see R.R. Dogonadze in N.S. Hush (Ed.), Reactions of Molecules at Electrodes, Wiley, London, 1971, p. 135; J.M. Hale in Hush op. cit., p. 229.
18 See e.g. R. Parsons and E. Passeron, J. Electroanal. Chem., 12 (1966) 524.
19 R.A. Marcus, J. Chem. Phys., 43 (1965) 679.
20 N.S. Hush, J. Chem. Phys., 28 (1958) 962.
21 N. Sutin, Ann. Rev. Nucl. Sci., 12 (1962) 285.